

A Cognitive Architecture That Produces Teleosemantic Content as Fossilized Wiring Between Independent Perception and Prediction Systems

Przemysław Siemion

Abstract

We propose a cognitive architecture in which perception and prediction are strictly independent systems with no shared vocabulary and no pre-established correspondence. A matching process, driven by prediction error, perceptual coverage, and available model capacity, searches for productive wirings between the two. Successful wirings are retained — fossilized — as the system’s ontological commitments. The routing is the ontology.

This design has consequences: the matching process requires a working substrate — a proto-ontology — without which perception and prediction cannot be connected. This intermediate structure is not designed into the system. It is the minimal precondition for the matching problem to be solvable. The implementation made it necessary before any theoretical argument.

The architecture re-discovers Millikan’s teleosemantics: meaning is constituted by the consumer’s success conditions, and fossilized wiring is its computational instantiation. Symbol grounding, in this framework, is not a relation between symbol and world — grounding comes from predictive success. The search goes over hypothesis spaces that are derived, not prescribed.

When the environment changes beyond an incumbent model’s capacity, the architecture supports model succession: competing predictive models, wired to the same perception stream, provide a basis for ontological shift as an inspectable architectural event. The system distinguishes between environments that require extending existing wirings and those that demand replacing them. What got wired can get re-wired — from within.

1 The Problem

In spite of enormous advances in practical AI applications, a decades-old problem persists: today’s best AI systems, even when presented with the most compelling evidence, have no mechanism for revising their ground truths without re-training.

This is not surprising. Ontological shift has been studied at length and described in depth, but never mechanized. The same unresolved element has been described by, inter alia, Kuhn in science, Quine in conceptual schemes, and Piaget in development.

Yet no convincing and realistic mechanism that could realize the ontological shift has been offered.

The problem cannot be resolved within the framework of Symbolic AI, nor within end-to-end learning, for seemingly different, but related reasons. In Symbolic AI, the ontology is given by the designer. In end-to-end training, ontology is dissolved into weights. Neither of these systems has the ability to reconfigure itself from the inside.

In this paper we propose a third architecture, in which perception is separate from predictive machinery. They share neither any a priori common vocabulary nor any pre-established correspondence. Biologically, the design principle is to minimize the innate correspondence. The cognitive effort goes into identifying mappings, or wirings, between perception and prediction that are successful in giving correct predictions. Ontological structure emerges as the residue of successful wiring. Grounding comes from predictive success. Ontological shift may be interpreted as radical re-wiring. As there is an explicit substrate, and an explicit process that governs the wiring, everything is inspectable.

The paper presents the architecture at the computational level in Marr’s sense: some components are conjectured, some have been implemented and tested, most of them have internal architecture that is encapsulated and can be explored independently. The architecture is implementable. A companion paper [?] demonstrates the architecture at work in a known minimal-case scenario. We want to stress that many rectifications of the architecture are coming directly from the implementation efforts.

2 The Independence Constraint and the Emergent Ontology

The first principle of the architecture is that perception and prediction are two independent systems, and that cognitive processes include, if not consist of, creating and governing the connections between the two. Any shared vocabulary, any tight structural coupling, should be avoided as it violates the principle.

Perception says nothing about what to expect next; prediction has no sense of “what” it predicts. It is the matching process that identifies the successful mapping between the two, and “successful” here has a very specific meaning: the prediction output, as interpreted by the wiring, correlates with the successive perception signals.

As perception produces categories in one vocabulary, and prediction operates in another, the search space is constrained. But the constraint comes from the vocabulary sizes of the two systems, not from an externally provided set of hypotheses. This is how the hypothesis-space problem of Bayesian cognitive science gets resolved by the architecture: the hypothesis space is effectively the space of possible wirings.

One consequence of positioning the matching process as the fundamental actor is that the process needs its working state. This issue became apparent when working on minimal implementations. The only honest way to deal with the matching process state is to make it an explicit component of the architecture. In retrospect we realize it is a catalyst substrate without which the perception and prediction matching

cannot happen. The initial state of the matching process is assumed to be blank, but existing. We call it “proto-ontology.”

The matching process identifies wiring with predictive properties, and fossilizes this wiring, effectively fusing the emergent ontology onto its proto-ontology substrate. And as the I/O elements of the prediction get wired to elements of perception, they, in a sense, assume *meaning*. But this meaning is emergent, operational, and up for revisions.

The Independence Constraint is stronger than modularity: modular systems share interface specifications. Here, no specification exists. The correspondence is the system’s problem to solve.

3 Architecture

The architecture is determined by one fundamental idea: interesting cognitive processes happen *between* perception and the predictive machinery, and are the only bridge between the two. Defining minimal meaningful perception and prediction components sets the stage.

Perception. A system that observes a world and produces discrete perceptual categories. In the simplest version it runs unsupervised. The architecture does not preclude top-down modulation from the matching process, but we defer opening this frontier in this presentation. In effect, the perception system does not need to have any internal anticipations. Its role is solely to resolve the sensory inputs onto a vocabulary that is discovered, not specified. It may have non-trivial internal architecture, but its essential role is to generate tokens from a slowly-evolving, context-specific vocabulary in response to external stimuli.

Prediction — a portfolio of predictive models. Each model is pre-trained in isolation. Therefore each model is an expert in its pre-training domain. In this sense, the models do have their internal ontologies. The crucial point is that the internal ontologies are never published. There is no labeling that would indicate what the model’s expertise is about. The models’ internals remain encapsulated. For this reason we call them “boxed models.”

Boxed models expose a homogeneous dictionary, a unified view of their inputs and outputs. They act as automata driven by signals from the matcher, and they respond by returning probability distributions over “next token.”

The distinction between inputs and outputs of a model remains hidden. A model can have logic based explicitly on some internal semantics — but these remain encapsulated and never affect the matcher. This is not just architectural strictness: being an input or output is a quality orthogonal to being predictively relevant.

The architecture does not preclude fine-tuning the models once they have been wired, but, as with the perception modulation, we consider it out of scope of this paper.

The matcher. The matcher does the cognitive heavy-lifting. It is the bridge between perception and the predictive models portfolio.

The matcher’s task is to establish a productive relationship between perception and the predictive machinery. It searches the space of possible wirings between the

perception dictionary and the models’ dictionaries. It has an internal measure of predictive error as its driver. The error function is generic: it simply assesses how probable the current perception-token is in the latest prediction from the currently wired model. But because the size and granularity of the dictionaries is arbitrary, two additional guiding signals are necessary. One is completeness: are all the world tokens mapped to prediction? Without this signal the system may have low prediction error because it quietly ignores some aspects of the perception stream. The other one is more subtle. It is the model’s capacity: it is not necessary that all the pads of the model are wired. In fact, unwired pads can be considered a hedge against a changing environment.

The search is oblivious in the sense that the matcher has no other intrinsic guidelines about which model to wire, and how to map the dictionaries. The architecture does not constrain the matcher to wire only one model to the perception.

The quality of a wiring reflects the balance between predictive accuracy, perceptual coverage, and available capacity. The predictive error is the main driver, the other two are supportive. What follows is that, by tuning the relative sensitivities to these signals, the architecture can approximate different cognitive profiles. For example, the reactivity to the presence of unwired perception tokens models the difference between “dogmatic” and “curious.”

The best wiring found is persisted, or “fossilized,” because it offers actual predictive power for the system as a whole. Competing wirings that perform reasonably may also be retained, as their models’ unwired capacity represents adaptive potential — a hedge against a changing environment. The fossilized wirings are all that the system has learned, and are the only place where any relation between perception and prediction is retained. A boxed model without wiring is an expert system. A perception stream without wiring to prediction is just that: a perception stream. It is our position that profound cognition begins at the point of connection.

Interface principle. The predictive model never sees raw perceptual data. It receives signals on its exposed pads from the matcher, but never knows how its predictions relate to the *outside*.¹

4 Grounding as Fossilized Wiring

Success of the matching process is understood as identifying a wiring between the vocabulary of perception and vocabularies of one, or more, predictive models, which results in predictions that are *useful*. We leave *useful* formally undefined here, precisely because the concept of usefulness is context-dependent. We loosely define a *good* wiring as providing a close approximation of the actual world-token stream.

Once a successful mapping has been established, a wiring exists: perceptual category X maps to model token Y. “X is interpreted as Y” not by representation, not by encoding, but by *routing*. The routing is the ontology. It has been discovered, it stems from predictive success, and it is *valuable*. As such, it gets preserved as a residual trace of a successful matching process.

¹The reader familiar with Stoppard’s *Dogg*’s language will recognize the scenario.

This way the architecture re-discovers the teleosemantics of Millikan: meaning is constituted by the success conditions of the consumer, not by resemblance or stipulation. In Millikan’s terms the predictive model is the consumer, and the wiring’s content is whatever has resulted in the consumer succeeding in predicting the future — as judged by the matcher. Fossilized wiring becomes a computational instantiation of the teleosemantic relation.

This in turn sheds light on the symbol grounding problem, as formulated by Harnad: how do symbols connect to the *world*? The architecture walks away from this problem: symbols do not relate directly to the world. Symbols search for their slots in the predictive machinery, and their fitness is measured by overall predictive success. Grounding is contextual, indirect, and it comes from predictive success.

Further: Fodor argues in *Hume Variations* that genuinely new concepts cannot be acquired — only triggered from an innate stock. The architecture half-agrees: predictive models are pre-existing, their internal vocabulary is not learned from the perception stream. But the model is not the concept. The concept is a successful application of a model: good predictive wiring to the world-context. In a departure from Fodor’s ideas, concepts appear to be operational units rather than just symbols.

There is also a flip side to the operational and emergent nature of grounding. And this flip side is the architecture’s greatest upside: what got wired, can get re-wired. We tacitly assume such architectures will adapt. At any point in time the perception stream may change, partially invalidating the existing predictive power, and triggering a search for a better, updated matching. A new wiring, to a different model in the portfolio, may be identified as the new champion and fossilized. No retraining. No updating of parameters from the outside. Just changing the routes. From within. Shifting the meanings in an almost Quinean sense. This is structurally different from Bayesian belief revision, as it is a change in the interpretation function, not in the probability distributions.

5 Model Competition and Succession

As stated, multiple predictive models can be wired to the same perception stream simultaneously. The models may be complementary, but they may also be competitive with one another. The matching process identifies a winning model, and fossilizes a wiring to it. But the winner status may be temporary. If the matching process also identifies one or more other wirings with reasonable predictive performance, they may be preserved, and their models wired. One reason for keeping these “contenders” could be their adaptive potential in the form of their model’s unwired pads.

Let us now describe what we call an “ontological crisis.” It is easy to model a situation in which the external stimuli and the perception stream change such that the current leading model stops being satisfactory. It may be that the perception dictionary remains unchanged, but the current wiring to the current leader’s internal logic does not correlate to perception any longer. It may be that the dictionary expands, presenting new perception tokens to wire. Or it could be both.

Heuristically, the third possibility is the most relevant — the world rarely changes its internal logic without new factors appearing. It also offers us the most vivid

scenario for model succession.

Assume there is a clear leader in the wiring space — a wiring that produces rock-solid predictions. But the wired model is at its capacity, all its pads have already been tapped. There are also one or more contenders, producing somewhat weaker predictions, but with some unwired tokens in the models’ dictionaries.

Now two things happen in conjunction: the predictions of the leading wiring start generating errors, and a new token appears in the perception dictionary.

The system needs to adapt, and the nature of the adaptation is obvious: include the new perception token in the predictive machinery and look for good correlations. But the current leader, its model being at the capacity, has no adaptive potential. The contenders, performing reasonably on the old domain and having spare pads to map, are the new hypothesis space to search through. If the system resolves the ontological crisis at all, it is one of the contenders, extended, that becomes the new lead.

There are many other ways, differing from the above scenario, that the succession can play out. And one has to be precise about distinguishing between the leading *wiring* and the leading *model*. It is a specific wiring to a specific model that makes a predictive unit. The companion paper [?] demonstrates a simple case where the succession happens between two wirings to the same model.

Coming back to the ontological crisis unfolding, there are two things to highlight. The first observation is that the system tries to deal with the crisis from within, and has non-zero chances of succeeding. The ontological shift becomes an observable event within the architecture.

The second observation is that by keeping contenders wired, by running competing models in synchrony, this ontological shift can be smoothed out and hedged. When ontological crisis happens, the incumbent wirings may still perform well on restricted domains, providing scaffolding that mitigates issues like catastrophic forgetting.

Figure 1: The three guiding signals in one frame: fossilized wiring (predictive error flows here), unmapped perception tokens (the itch), and unwired model capacity (adaptive potential).

In the course of experiments with implementing this architecture we found it necessary to clearly articulate the three signals which drive the system. The predictive error is the obvious one. But it turns out to be insufficient on its own. When new token types begin to show up in the perception stream, it may well be that the system can keep the predictive error low by quietly ignoring them. The presence of unwired perceptions needs to be a signal in itself. We call it the “cognitive itch,” and the low reactivity to the itch is a sort of blindness. Sudden occurrence of predictive error calls for a different reaction, depending on the presence or absence of unwired perception tokens. Coincidence of predictive error and new perceptions calls for extending the current wirings, seeking to incorporate the new signals from the world. Rise of the predictive error in absence of new perceptions calls for radical rewiring and, concurrently, should trigger an active search for any new perception tokens — a frontier we have deferred in Section 3. Likewise, the “adaptive potential” of the unwired pads of a model does not affect the predictive error. But a wiring to a

model with some adaptive potential inherits this potential, which makes this wiring valuable for reasons that are different from its current predictive precision. Any wirings portfolio management strategy that wants to account for adaptability needs to be aware of the adaptive potential on top of the current predictive performance.

The origin and pre-selection of the predictive models portfolio, the bootstrap under weak initial correlations, and the evolution of models through fine-tuning once wired, are important practical concerns that do not affect the architectural principles. We touch upon these issues in Section 8.

6 Computational Landscape

Helmholtz machine. The Helmholtz machine [?] calls for two complementary but separate regimes of operation: a recognition model and a generative model. It is clear that a full cognitive architecture must cover both inference and exploitation. In this paper we are focused on fleshing out some details of a possible inference process. The fossilization does provide a point of support for a potential switch between the operational modes.

Bayesian program learning. Lake, Salakhutdinov, and Tenenbaum [?] learn compositional generative programs from few examples. BPL shares our emphasis on structured representations and sparse-data learning. Our program is more ambitious in the sense that the hypothesis space is not directly prescribed, but is much more modest in the sense that at this point we are concerned with signal routing rather than signal formatting. We feel that many detailed aspects of the architecture will benefit from probabilistic treatment once the foundations are stable.

Predictive processing. The free energy framework [?] and the broader predictive processing program [?] share the principle that prediction error is the fundamental cognitive signal. These architectures are rich, hierarchical, and inspiring. Their focus is on the exploitation. Our architecture is focused on inference. Prediction error evaluates candidate wirings between systems rather than refining signal processing. Our program seems complementary. We are deferring the frontier of how top-down modulation fits into the architecture.

Global Workspace Theory. Baars [?] and its computational extensions [?] propose a shared workspace through which specialized processors broadcast information. Our perception stream bears a surface resemblance to this workspace. The difference: in GWT, the workspace is a communication channel from which processors deposit and retrieve information. In our architecture, the perception stream carries no meaning until the matching process gives it one. It is raw material, not a message board.

7 Theoretical Positioning

Several behaviors of the architecture correspond tightly with established ideas in cognitive science. These correspondences are double-edged: they demonstrate possible mechanistic realizations, but at the same time bring to light their mechanical constraints.

Millikan. As already stated in Section 4, the ontology-as-wiring resulting from an interactive process is the architecture’s version of Millikan’s teleosemantics [?]. The wiring process is constructive, offering some insight into how teleosemantic relations can actually arise. Conversely, Millikan’s theory tells us how to think about the wirings, embedding the architecture into the body of core cognitive science.

Gopnik / Carey. Looked at through the lens of Gopnik and Carey’s theory-theory [?, ?], a specific wiring between a specific model and the perception becomes “a theory about the world.” The succession becomes the theory change. The “hypothesis testing” is realized by an explicit, mechanistic process. The conceptual repertoire, coming from available models and possible wirings, is still pre-determined, but indirectly. And the architecture offers an interesting staging to the repertoire extensions: learn new models, find new applications. The succession dynamics described in Section 5 provide a concrete mechanism for what Gopnik and Carey describe as theory change.

Fodor. A wired model, an established theory of the world, generating predictions by driving an automaton — this is as close to Fodor’s vision [?] as a simple architecture can get. But the wirings are not externally provided, and finding them makes “concepts” operational, not static symbols.

8 Open Problems

From framework to theory. The architecture, as implemented in a minimalistic setup described in the companion paper, has generated results that surprised its author. Whether it can do the same for an independent reader is the threshold between framework and theory. This paper is an invitation to test that.

Conditions for failure. It is possible, and actually probable, that neither the matching process, nor the bootstrap process, are successful. There could be worlds to which the simplistic mind of the architecture cannot meaningfully relate, given a specific initial model portfolio. We can only say that the overarching goal of the architecture is not to eliminate the need for the innate, but to search for ways to minimize it, to find a lower bound for what is known a priori which still allows for a successful bootstrap.

Perception granularity. The architecture as presented assumes unsupervised perception produces categories at the right granularity for available models. This assumption is not categorical, it is tactical and allows the presentation to be tractable. Sitting at the crossroads of prediction and perception, cognition can and should be detecting and signalling the success or the lack of it both up and down. We hope to follow this trajectory, but only after consolidating the current scope.

Inter-model consistency and complementarity. We have only hinted at inter-model consistency checks and their possible role in mitigating the risks of model succession, and we have not discussed complementary wirings at all. We consider these to be the scope of future work, where actual modelling can give enough substance for the proper treatment.

Commitment under partial information. A wiring that predicts well locally may not generalize. The architecture’s response to incorrect fossilization is not prevention

but correction — succession provides the mechanism for displacing an incumbent wiring when its predictive power degrades. The quality of the correction depends on the contender pool and the signal structure. In fact, the available perception stream is always finite, and the prediction almost certainly is never perfect. So a valid question is: under what criteria is the matching process actually resolved? This is the finite-horizon decision problem at the heart of real-time operation. It is our view that, because such criteria are pragmatic in nature, it would be perilous to tackle this problem before studying some working simulations.

Temporal alignment. The current implementation assumes one-step temporal alignment between prediction and perception. Relaxing this assumption introduces a temporal search dimension that compounds the matching. We note the problem and defer it.

The bootstrap and model fine-tuning. Another valid question is the origin of the predictive machinery. We could avoid this question altogether, but let us share our experimentation-driven conjecture: the architecture does require a pool of pre-existing predictive models from which candidates are pre-selected based on initial fit to the perception stream. It is not guaranteed that a strongly correlating wiring can be found with the initial model portfolio. But even in case of weak initial correlations the best-performing models and wirings are selected, and fine-tuned through a bootstrap period. If the bootstrap process is successful, the system eventually achieves a degree of predictive success even when no initial model fits immediately. It is not a first-order architectural issue and is not worked out in detail. Actual simulations are the proper way to work out correct operational details.

9 Conclusion

The architecture, as presented here, is a result of trying to implement a high-level idea in actual code. At many points in the process we felt our hand being forced. The independence constraint has forced a matching problem. The matching problem has forced an intermediate substrate. Successful matching has produced a problem of retaining its outcome — hence fossilization. This gives us a sense of confidence in the architecture, but a specific one: what works, works, maybe at the price of missing some broader perspective. The perspective may be narrowing, but ontological shift becomes inspectable, which is a deal we take.

Final Notes

“Fossilized wiring” as a term openly borrows from Selinker’s “fossilized errors” [?]. However, we would like to note that here “fossilization” acquires a positive spin, which it deserves.

The architecture is verifiably productive in the sense that we know at least one theatrical play, *Dogg’s Hamlet* by Tom Stoppard [?], that hinges on its very principles. Stoppard wrote it in 1979, which is a notable foresight.

The architecture was developed in conversation with large language models, which served as both thinking partners and experimental substrates (as they were

quick to realize).

We have not been able to escape the biological and anthropocentric undertones. “Minimizing the innate” is, of course, self-reflective, and it is the original fuel for our effort.

References

- [1] [Companion paper — link to be added upon publication.]
- [2] Baars, B. J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press.
- [3] Carey, S. (2009). *The Origin of Concepts*. Oxford University Press.
- [4] Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204.
- [5] Dayan, P., Hinton, G. E., Neal, R. M., & Zemel, R. S. (1995). The Helmholtz machine. *Neural Computation*, 7(5), 889–904.
- [6] Dehaene, S., & Changeux, J. P. (2011). Experimental and theoretical approaches to conscious processing. *Neuron*, 70(2), 200–227.
- [7] Fodor, J. A. (2004). *Hume Variations*. Oxford University Press.
- [8] Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.
- [9] Gopnik, A., & Meltzoff, A. N. (1997). *Words, Thoughts, and Theories*. MIT Press.
- [10] Harnad, S. (1990). The symbol grounding problem. *Physica D*, 42(1–3), 335–346.
- [11] Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. University of Chicago Press.
- [12] Lake, B. M., Salakhutdinov, R., & Tenenbaum, J. B. (2015). Human-level concept learning through probabilistic program induction. *Science*, 350(6266), 1332–1338.
- [13] Marr, D. (1982). *Vision*. W. H. Freeman.
- [14] Millikan, R. G. (1984). *Language, Thought, and Other Biological Categories*. MIT Press.
- [15] Piaget, J. (1952). *The Origins of Intelligence in Children*. International Universities Press.
- [16] Quine, W. V. O. (1951). Two dogmas of empiricism. *Philosophical Review*, 60(1), 20–43.

- [17] Selinker, L. (1972). Interlanguage. *International Review of Applied Linguistics*, 10(3), 209–231.
- [18] Stoppard, T. (1979). *Dogg's Hamlet, Cahoot's Macbeth*. Samuel French.